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Applied Data Mining

Assignment 2 – Screenshots and Explanations

1. Lines 1 to 18

The screenshot displays the Posit Cloud workspace for 'Module3'. The main editor shows R code for data analysis, and the console shows the execution output.

Code (Lines 1 to 18):

```
1 ### Table 14.4
2 library(arules)
3 fp.df <- read.csv("Faceplate.csv")
4
5 # remove first column and convert to matrix
6 fp.mat <- as.matrix(fp.df[, -1])
7
8 # convert the binary incidence matrix into a transactions database
9 fp.trans <- as(fp.mat, "transactions")
10 inspect(fp.trans)
11
12 ## get rules
13 # when running apriori(), include the minimum support, minimum confidence, and target
14 # as arguments.
15 rules <- apriori(fp.trans, parameter = list(supp = 0.2, conf = 0.5, target = "rules"))
16
17 # inspect the first six rules, sorted by their lift
18 inspect(head(sort(rules, by = "lift"), n = 6))
```

Environment:

Object	Class	Size
fp.df	data.frame	10 obs. of 7 variables
fp.mat	matrix	int [1:10, 1:6] 1 0 0 1 1 0 1 1 0 ...
fp.trans	transactions	Formal class transactions
rules	rules	Formal class rules

Files:

Name	Size	Modified
..		
.Rhistory	0 B	Mar 28, 2026, 5:56 PM
project.Rproj	205 B	Mar 28, 2026, 5:56 PM
Faceplate.csv	199 B	Mar 28, 2026, 5:58 PM
Ch14.R	1.9 KB	Mar 28, 2026, 5:58 PM

Console Output:

```
R - R 4.5.3 - /cloud/project/
set transactions ... [6 item(s), 10 transaction(s)] done [0.00s].
sorting and recoding items ... [5 item(s)] done [0.00s].
creating transaction tree ... done [0.00s].
checking subsets of size 1 2 3 done [0.00s].
writing ... [18 rule(s)] done [0.00s].
creating S4 object ... done [0.00s].
>
> # inspect the first six rules, sorted by their lift
> inspect(head(sort(rules, by = "lift"), n = 6))
lhs      rhs      support confidence coverage lift      count
[1] {Red, White} => {Green} 0.2      0.5      0.4      2.500000 2
[2] {Green} => {Red} 0.2      1.0      0.2      1.666667 2
[3] {White, Green} => {Red} 0.2      1.0      0.2      1.666667 2
[4] {Orange} => {White} 0.2      1.0      0.2      1.428571 2
[5] {Green} => {White} 0.2      1.0      0.2      1.428571 2
[6] {Red, Green} => {White} 0.2      1.0      0.2      1.428571 2
>
```

```

> #### Table 14.4
> library(arules)
> fp.df <- read.csv("Faceplate.csv")
>
> # remove first column and convert to matrix
> fp.mat <- as.matrix(fp.df[, -1])
>
> # convert the binary incidence matrix into a transactions database
> fp.trans <- as(fp.mat, "transactions")
> inspect(fp.trans)
      items
[1] {Red, White, Green}
[2] {White, Orange}
[3] {White, Blue}
[4] {Red, White, Orange}
[5] {Red, Blue}
[6] {White, Blue}
[7] {Red, Blue}
[8] {Red, White, Blue, Green}
[9] {Red, White, Blue}
[10] {Yellow}
>
> ## get rules
> # when running apriori(), include the minimum support, minimum confidence, and target
> # as arguments.
> rules <- apriori(fp.trans, parameter = list(supp = 0.2, conf = 0.5, target = "rules"))
Apriori

Parameter specification:
 confidence minval  smax  arem   aval originalSupport  maxtime support minlen maxlen target  ext
      0.5      0.1    1 none FALSE             TRUE         5     0.2     1    10 rules TRUE

Algorithmic control:
 filter tree heap memopt load sort verbose
  0.1 TRUE TRUE  FALSE TRUE    2     TRUE

Absolute minimum support count: 2

set item appearances ...[0 item(s)] done [0.00s].
set transactions ...[6 item(s), 10 transaction(s)] done [0.00s].
sorting and recoding items ... [5 item(s)] done [0.00s].
creating transaction tree ... done [0.00s].
checking subsets of size 1 2 3 done [0.00s].
writing ... [18 rule(s)] done [0.00s].
creating S4 object ... done [0.00s].
>
> # inspect the first six rules, sorted by their lift
> inspect(head(sort(rules, by = "lift"), n = 6))
      lhs              rhs  support confidence coverage lift      count
[1] {Red, White} => {Green} 0.2      0.5      0.4      2.500000 2
[2] {Green}      => {Red}   0.2      1.0      0.2      1.666667 2
[3] {White, Green} => {Red}  0.2      1.0      0.2      1.666667 2
[4] {Orange}     => {White} 0.2      1.0      0.2      1.428571 2
[5] {Green}      => {White} 0.2      1.0      0.2      1.428571 2
[6] {Red, Green} => {White} 0.2      1.0      0.2      1.428571 2

```

Explanation:

The dataset was first loaded from a CSV file and stored as a data frame (fp.df). The first column, which contained non-relevant identifiers, was removed and the remaining data was converted into a binary matrix (fp.mat) where 1 indicates the presence of an item and 0 indicates its absence. The matrix was converted to a transactions object (fp.trans) which served as the necessary format for association rule mining with the arules package. The Apriori algorithm was applied with a minimum support of 0.2 and a minimum confidence of 0.5 to generate association rules. A total of 18 rules were produced and the strongest rules were identified by sorting them based on lift. The rules establish item relationships which show how certain items presence leads to increased chances of other items appearing in the same transaction.

The screenshot displays the Posit Cloud workspace for 'Module3'. The main editor shows R code for inspecting and filtering association rules. The console output lists 18 rules with their support, confidence, coverage, lift, and count. The environment pane on the right shows the loaded objects: fp.df (10 obs. of 7 variables), fp.mat (int [1:10, 1:6] 1 0 0 1 1 0 1 1 0 ...), fp.trans (Formal class transactions), rules (Formal class rules), and rules.tbl (18 obs. of 8 variables). The files pane at the bottom shows the project structure, including .Rhistory, project.Rproj, Faceplate.csv, and Ch14.R.

```
## Table 14.6
rules.tbl <- inspect(rules)
rules.tbl[rules.tbl$support >= 0.04 & rules.tbl$confidence >= 0.7,]
```

lhs	rhs	support	confidence	coverage	lift	count
[1] {}	=> {Blue}	0.6	0.6000000	1.0	1.0000000	6
[2] {}	=> {Red}	0.6	0.6000000	1.0	1.0000000	6
[3] {}	=> {White}	0.7	0.7000000	1.0	1.0000000	7
[4] {Orange}	=> {White}	0.2	1.0000000	0.2	1.4285714	2
[5] {Green}	=> {Red}	0.2	1.0000000	0.2	1.6666667	2
[6] {Green}	=> {White}	0.2	1.0000000	0.2	1.4285714	2
[7] {Blue}	=> {Red}	0.4	0.6666667	0.6	1.1111111	4
[8] {Red}	=> {Blue}	0.4	0.6666667	0.6	1.1111111	4
[9] {Blue}	=> {White}	0.4	0.6666667	0.6	0.9523810	4
[10] {White}	=> {Blue}	0.4	0.5714286	0.7	0.9523810	4
[11] {Red}	=> {White}	0.4	0.6666667	0.6	0.9523810	4
[12] {White}	=> {Red}	0.4	0.5714286	0.7	0.9523810	4
[13] {Red, Green}	=> {White}	0.2	1.0000000	0.2	1.4285714	2
[14] {White, Green}	=> {Red}	0.2	1.0000000	0.2	1.6666667	2
[15] {Red, White}	=> {Green}	0.2	0.5000000	0.4	2.5000000	2
[16] {Red, Blue}	=> {White}	0.2	0.5000000	0.4	0.7142857	2
[17] {White, Blue}	=> {Red}	0.2	0.5000000	0.4	0.8333333	2
[18] {Red, White}	=> {Blue}	0.2	0.5000000	0.4	0.8333333	2

```
> rules.tbl[rules.tbl$support >= 0.04 & rules.tbl$confidence >= 0.7,]
```

lhs	rhs	support	confidence	coverage	lift	count
[3] {}	=> {White}	0.7	0.7	1.0	1.000000	7
[4] {Orange}	=> {White}	0.2	1.0	0.2	1.428571	2
[5] {Green}	=> {Red}	0.2	1.0	0.2	1.666667	2
[6] {Green}	=> {White}	0.2	1.0	0.2	1.428571	2
[13] {Red, Green}	=> {White}	0.2	1.0	0.2	1.428571	2
[14] {White, Green}	=> {Red}	0.2	1.0	0.2	1.666667	2

Lines 22 to 25

The screenshot displays the Posit Cloud workspace for 'Module3'. The R script editor shows the following code:

```
21  
22 ##### Table 14.6  
23  
24 rules.tbl <- inspect(rules)  
25 rules.tbl[rules.tbl$support >= 0.04 & rules.tbl$confidence >= 0.7,]  
26
```

The console output shows the results of the `inspect(rules)` function, listing 18 rules with their left-hand side (lhs), right-hand side (rhs), support, confidence, coverage, lift, and count.

lhs	rhs	support	confidence	coverage	lift	count
[1] {}	=> {Blue}	0.6	0.6000000	1.0	1.0000000	6
[2] {}	=> {Red}	0.6	0.6000000	1.0	1.0000000	6
[3] {}	=> {White}	0.7	0.7000000	1.0	1.0000000	7
[4] {Orange}	=> {White}	0.2	1.0000000	0.2	1.4285714	2
[5] {Green}	=> {Red}	0.2	1.0000000	0.2	1.6666667	2
[6] {Green}	=> {White}	0.2	1.0000000	0.2	1.4285714	2
[7] {Blue}	=> {Red}	0.4	0.6666667	0.6	1.1111111	4
[8] {Red}	=> {Blue}	0.4	0.6666667	0.6	1.1111111	4
[9] {Blue}	=> {White}	0.4	0.6666667	0.6	0.9523810	4
[10] {White}	=> {Blue}	0.4	0.5714286	0.7	0.9523810	4
[11] {Red}	=> {White}	0.4	0.6666667	0.6	0.9523810	4
[12] {White}	=> {Red}	0.4	0.5714286	0.7	0.9523810	4
[13] {Red, Green}	=> {White}	0.2	1.0000000	0.2	1.4285714	2
[14] {White, Green}	=> {Red}	0.2	1.0000000	0.2	1.6666667	2
[15] {Red, White}	=> {Green}	0.2	0.5000000	0.4	2.5000000	2
[16] {Red, Blue}	=> {White}	0.2	0.5000000	0.4	0.7142857	2
[17] {White, Blue}	=> {Red}	0.2	0.5000000	0.4	0.8333333	2
[18] {Red, White}	=> {Blue}	0.2	0.5000000	0.4	0.8333333	2

```
> ##### Table 14.6
```

```
>
```

```
> rules.tbl <- inspect(rules)
```

	lhs	rhs	support	confidence	coverage	lift	count
[1]	{}	=> {Blue}	0.6	0.6000000	1.0	1.0000000	6
[2]	{}	=> {Red}	0.6	0.6000000	1.0	1.0000000	6
[3]	{}	=> {White}	0.7	0.7000000	1.0	1.0000000	7
[4]	{Orange}	=> {White}	0.2	1.0000000	0.2	1.4285714	2
[5]	{Green}	=> {Red}	0.2	1.0000000	0.2	1.6666667	2
[6]	{Green}	=> {White}	0.2	1.0000000	0.2	1.4285714	2
[7]	{Blue}	=> {Red}	0.4	0.6666667	0.6	1.1111111	4
[8]	{Red}	=> {Blue}	0.4	0.6666667	0.6	1.1111111	4
[9]	{Blue}	=> {White}	0.4	0.6666667	0.6	0.9523810	4
[10]	{White}	=> {Blue}	0.4	0.5714286	0.7	0.9523810	4
[11]	{Red}	=> {White}	0.4	0.6666667	0.6	0.9523810	4
[12]	{White}	=> {Red}	0.4	0.5714286	0.7	0.9523810	4
[13]	{Red, Green}	=> {White}	0.2	1.0000000	0.2	1.4285714	2
[14]	{White, Green}	=> {Red}	0.2	1.0000000	0.2	1.6666667	2
[15]	{Red, White}	=> {Green}	0.2	0.5000000	0.4	2.5000000	2
[16]	{Red, Blue}	=> {White}	0.2	0.5000000	0.4	0.7142857	2
[17]	{White, Blue}	=> {Red}	0.2	0.5000000	0.4	0.8333333	2
[18]	{Red, White}	=> {Blue}	0.2	0.5000000	0.4	0.8333333	2

```
> rules.tbl[rules.tbl$support >= 0.04 & rules.tbl$confidence >= 0.7,]
```

	lhs	rhs	support	confidence	coverage	lift	count
[3]	{}	=> {White}	0.7	0.7	1.0	1.000000	7
[4]	{Orange}	=> {White}	0.2	1.0	0.2	1.428571	2
[5]	{Green}	=> {Red}	0.2	1.0	0.2	1.666667	2
[6]	{Green}	=> {White}	0.2	1.0	0.2	1.428571	2
[13]	{Red, Green}	=> {White}	0.2	1.0	0.2	1.428571	2
[14]	{White, Green}	=> {Red}	0.2	1.0	0.2	1.666667	2

Explanation:

The Apriori algorithm rules were transformed into the rules.tbl data frame through the inspect() function which provides users with better viewing and filtering capabilities. The rules contain support and confidence and coverage and lift, which enable users to assess the strength and value of their rule connections.

Applied a filtering condition to extract important rules which required support values of 0.04 or higher and confidence values of 0.7 or higher to be included in the analysis. The strong associations were identified through the process which resulted in a limited number of rules being selected. The rules demonstrate high confidence relationships which show that certain items always appear together with a confidence level of 1.0, which establishes strong dataset patterns. Green presence always results in Red and White presence, while Orange presence always occurs together with White.